

Ahmed Kheder

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Profile

Computer and Systems Engineering graduate from Ain Shams University, ranked in the top 9% of my class. I focus on LLM agents and applied machine learning, and I enjoy the whole cycle: designing the system, building it, and measuring whether it actually works. Comfortable moving between research ideas and clean, working code.

Education

Ain Shams University Cairo, Egypt
BSc in Computer and Systems Engineering Sep 2021 – Jul 2026

- **GPA:** 3.57/4.00 | 15th of 170 (Top 9%)
- **Key Modules:** Machine Learning, Deep Learning, Operating Systems, Database Systems, Data Structures & Algorithms, Advanced Programming, Artificial Intelligence, Fundamentals of Big-Data Analytics.

Experience

Nile University Giza, Egypt
Research Intern Jul 2026 – Sep 2026

- Researched real-time ECG forecasting, exploring reservoir computing, next-generation reservoir computing (NG-RC), Transformers, and time-series foundation models.

Projects

AI-Driven Functional Verification for Digital Designs | *Python, LangGraph, LangChain, MCP*

- Built VAI, an autonomous LLM agent (LangGraph) that automatically tests chip designs, reaching full coverage on 4 of 5 open-source designs and closing 84% of the testing gaps left by the traditional method.
- Reached full coverage in 7 minutes on average, compared with 60-minute traditional runs that still fell short, at an average LLM cost of ~\$0.30 per design
- Wrote a layered system prompt defining the agent's role, constraints, tool-usage rules, code-generation conventions, and error-recovery behavior
- Designed an MCP-style tool layer with structured outputs, separating the model's generated code from tool-invocation schemas via runtime state injection
- Implemented tiered context management with per-tier token budgets and agent self-summarization of over-budget context, persisted across restarts via checkpointing
- Built a persistent cross-run memory: a deduplicated strategy store plus a project-scoped relational knowledge base for retrieving past solutions
- Designed the VAI score, a novel composite metric for benchmarking AI-driven verification

Real-Time ECG Waveform Forecasting | *Python, PyTorch, Chronos, Transformer, Fine Tuning*

- Built a real-time system forecasting the next second of ECG by dividing the signal into beat timing, beat type, beat shape, and a slow low-frequency component, forecast by a zero-shot Chronos model, a beat-type classifier, and fine-tuned Chronos-Bolt models
- Achieved NMAE of 0.018 and MSE of 0.10 on patients never seen in training
- Designed a new metric scoring each forecast beat separately on timing, height, and shape, since standard error metrics blend these errors and can reward flat, smeared forecasts; the typical forecast placed beats within 15 ms (about 2% of a heartbeat) at 76% of their true height, with 27% shape error
- Outperformed the FADE paper (2025) on normal rhythm by: 14% lower NMAE, 7% lower MAE, and beats at 81% vs. 72% of true height with 27% lower shape error

Motor Imagery Classification | *Python, PyTorch, TensorFlow, MNE*

[GitHub Repository](#) 

- Built an automated MNE-Python preprocessing pipeline for Muse-S EEG: band-pass and notch filtering, ICA artifact removal, CAR, and z-score normalization
- Applied offline augmentation (Gaussian noise, temporal shifts, window jitter) and packaged data into efficient PyTorch Dataset/DataLoader pipelines
- Implemented and fine-tuned EEGNet-8-2 CNN and Bi-LSTM architectures in PyTorch/TensorFlow with reproducible training and checkpointing

San Francisco Crime Classification | *Python, Scikit-learn, LightGBM*

[GitHub Repository](#) 

- Engineered temporal and spatial features, removed outliers and duplicates, and encoded/scaled inputs for multi-class classification
- Trained and tuned Random Forest, LightGBM, and Logistic Regression models, benchmarked with multi-class ROC curves and spatial visualizations

Professional Development

Machine Learning Specialization | *DeepLearning.AI & Stanford University* | supervised and unsupervised learning, reinforcement learning

AI, Data Science & Machine Learning Diploma | *WE Telecom Egypt* | data cleaning, feature engineering, model tuning, deep learning with TensorFlow (CNNs, RNNs)

Technical Skills

Languages: Python, C++, SQL

LLMs & Agents: LangGraph, LangChain, ReAct Agents, Prompt Engineering, Tool Calling / MCP, Context & Memory Management

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, MNE-Python

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Tools: Git/GitHub, MySQL, MongoDB